

Serpenoid robot

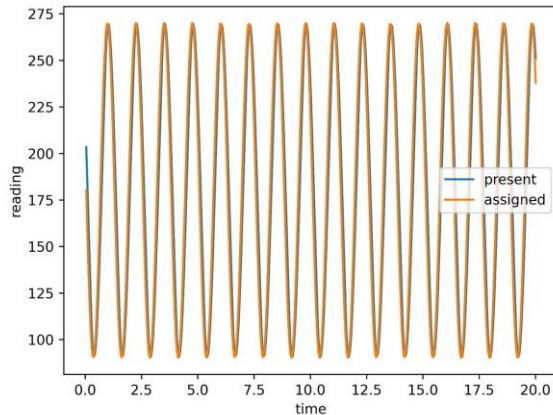
Primary testbed design and data acquisition

- Dynamixel MX-64R's are daisy chained to each other and connected to a central computer via a USB-RS485 converter
- Python and MATLAB was used for programming and data acquisition
- Motor position, current, supply voltage and temperature are acquired.

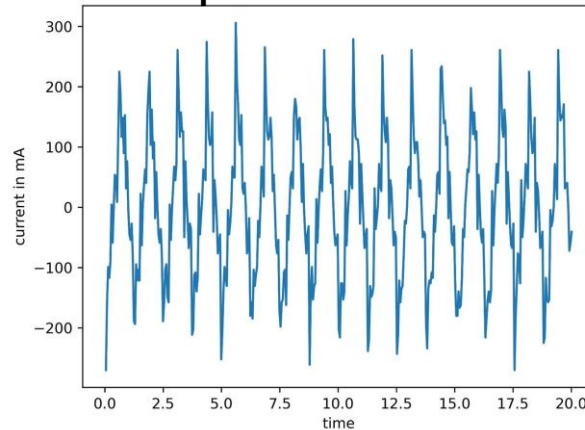


Source: https://emanual.robotis.com/assets/images/parts/interface/u2d2_01.png

Position control of DC servo



Variation of current during position control



Joints were 3D-printed, attached to the motor and tested



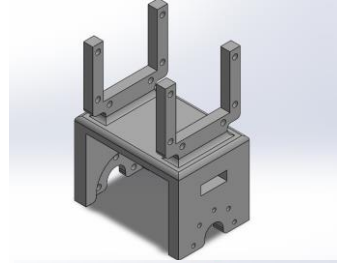
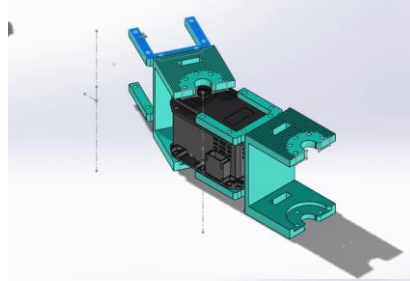
Iterative design assembly and testing

Design consists of resin 3D printed ABS modules with parallel joint axes, targeting the snake's movement in 2 dimensions.

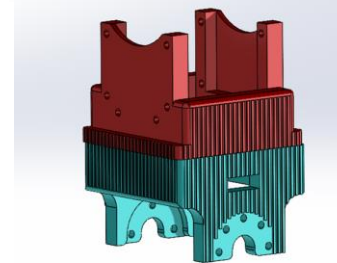
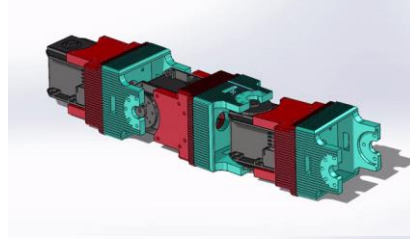
CAD-Assembly

Connector Module

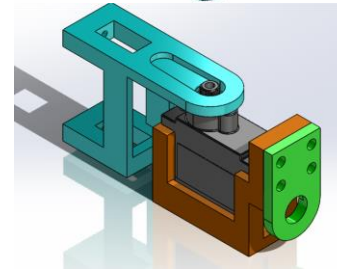
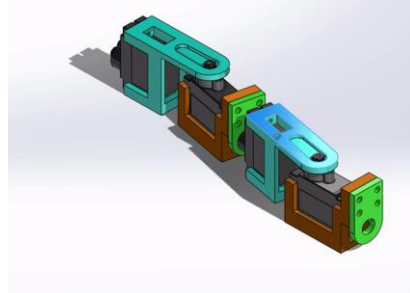
Design 1
(current)



Design 2
(in progress)

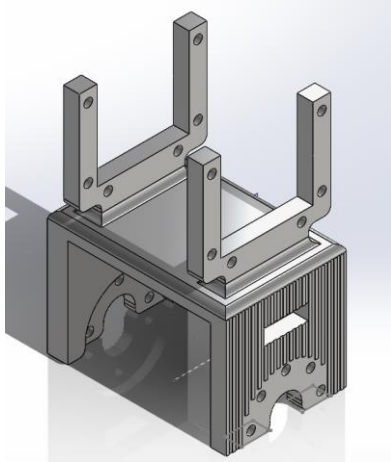


Design 3
(under development)

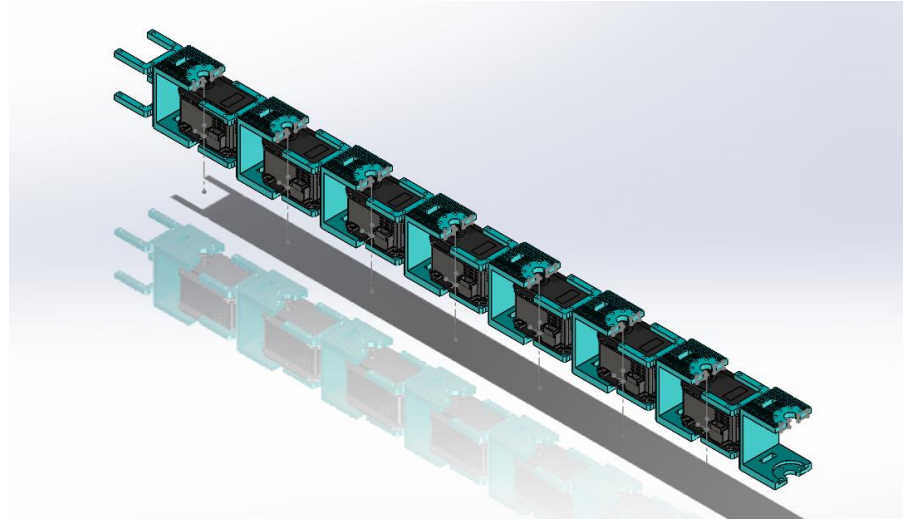


Details of design 1

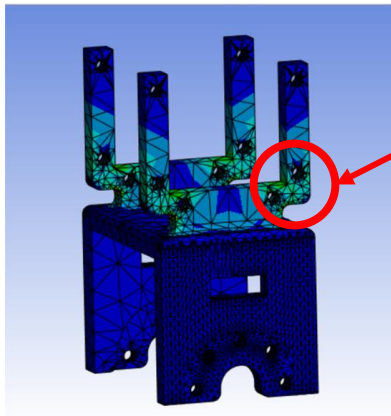
Iteration 1



Link

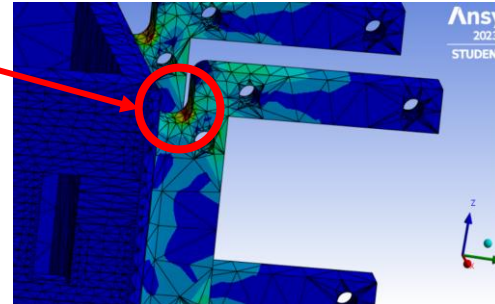


CAD assembly



**Weak
region**

Ansys stress test*

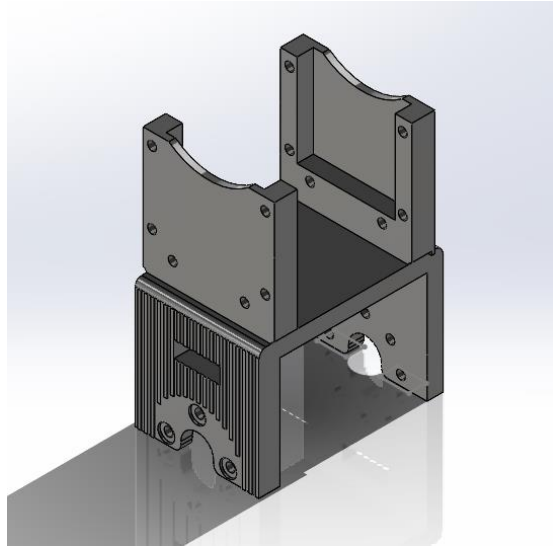


*Moment force of 6 N-m is applied
on the linke surfaces

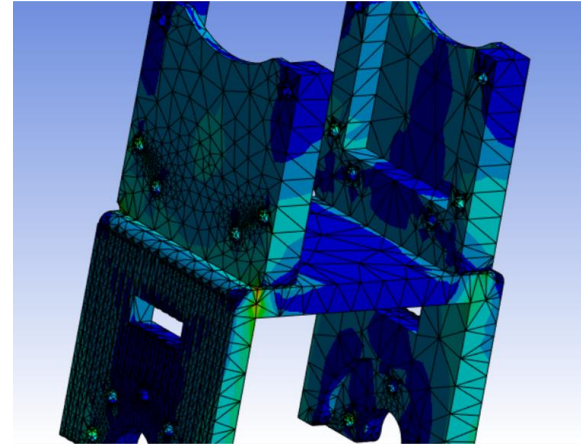
Corrections to design 1

Iteration 2

Considering the ansys study the links are modified in order to make links more robust.



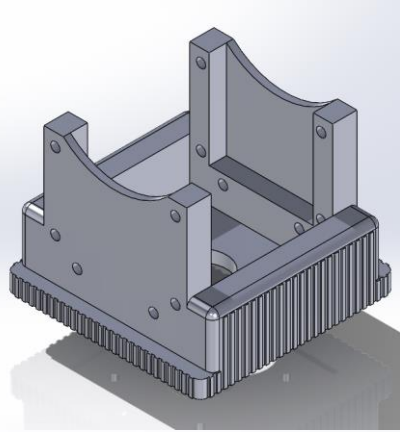
Modified Link



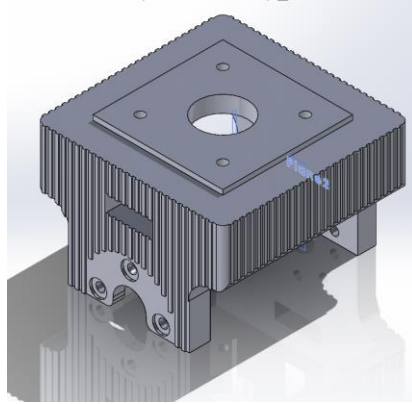
Ansys test on
modified link*

*Moment force of 6 N-m is applied
on the linke surfaces

Design 2



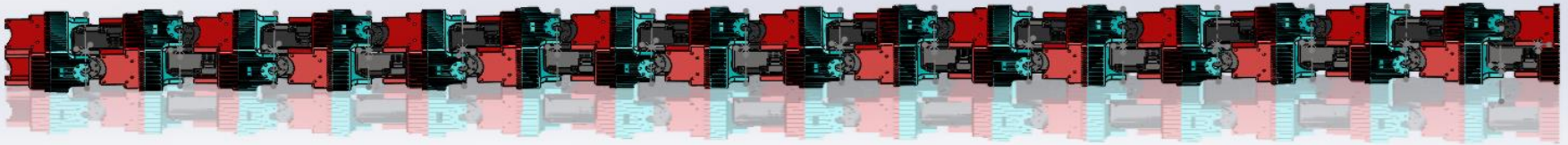
Link a



Link b

Links a and b are designed such that they can be arranged in 2 different states:

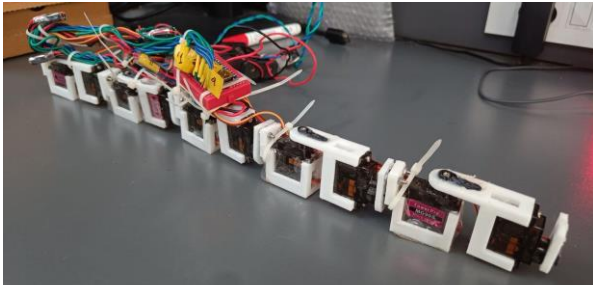
- Purely lateral
- Lateral and Dorsal



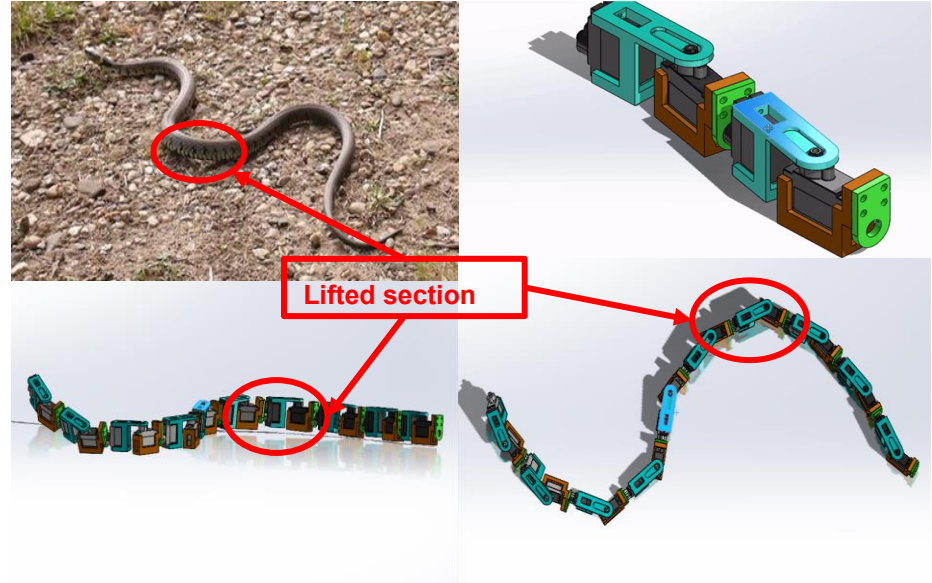
The CAD assembly comprise of total 16 joints, half is for forming a plane wave and other alternate joints are for lifting the body.

Design 3

- Each link comprise of 2 DoF
- One for the generation of **Lateral movement** and the other is for generating a **twist**



Low cost hardware

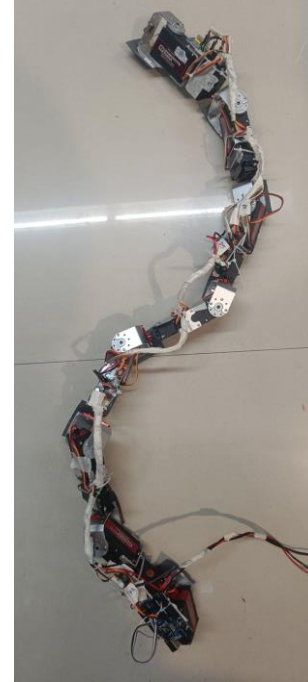


Robust hardware under development

Various prototypes



Design 1



Design 3

Implementation of concertina motion

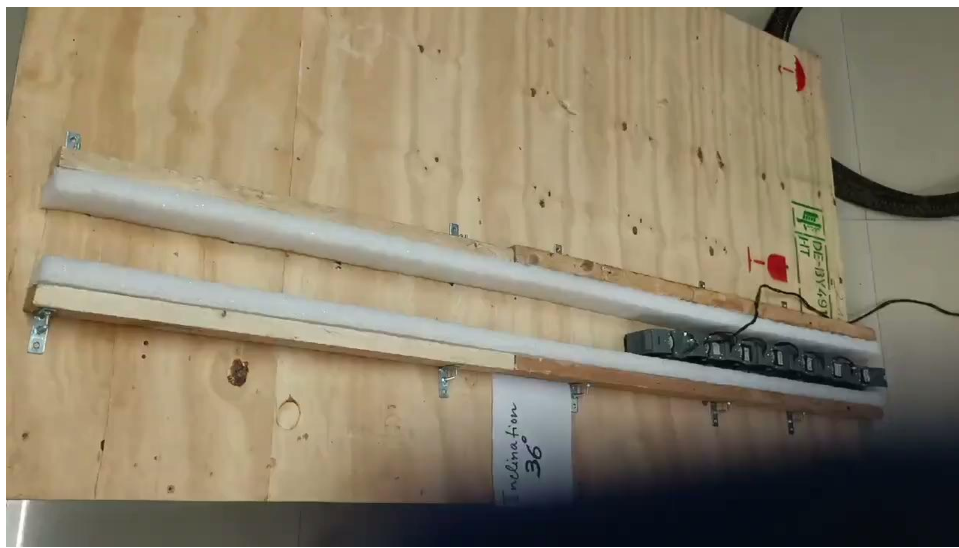
Concertina Motion in Real Snakes



Fig: Concertina in different channel width. (<https://youtu.be/a9y-C7LDpq0?si=hf8veaifxUASSdFx>)



- The mode of locomotion in limbless vertebrates while traversing tight spaces.
- This mode of locomotion is composed of mainly two phases.
 - Anchoring part the body to the sidewalls and extending the remaining part of the body forward.
 - Anchoring the extended part to the walls and contracting the remaining part.
- The motion uses the friction of the sidewalls and can also enhance the net frictional forces acting on the body by pushing against the wall.



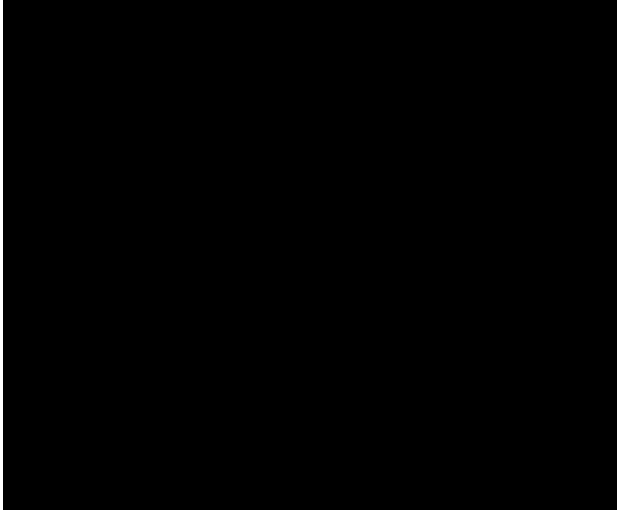
Videos demonstrating the concertina motion in a serpentine robot with 8 links climbing an inclination of 36 degrees.



Video demonstrating the concertina motion in a channel placed flat on ground

Implementation of lateral undulation

Proposed control algorithm for sidewinding



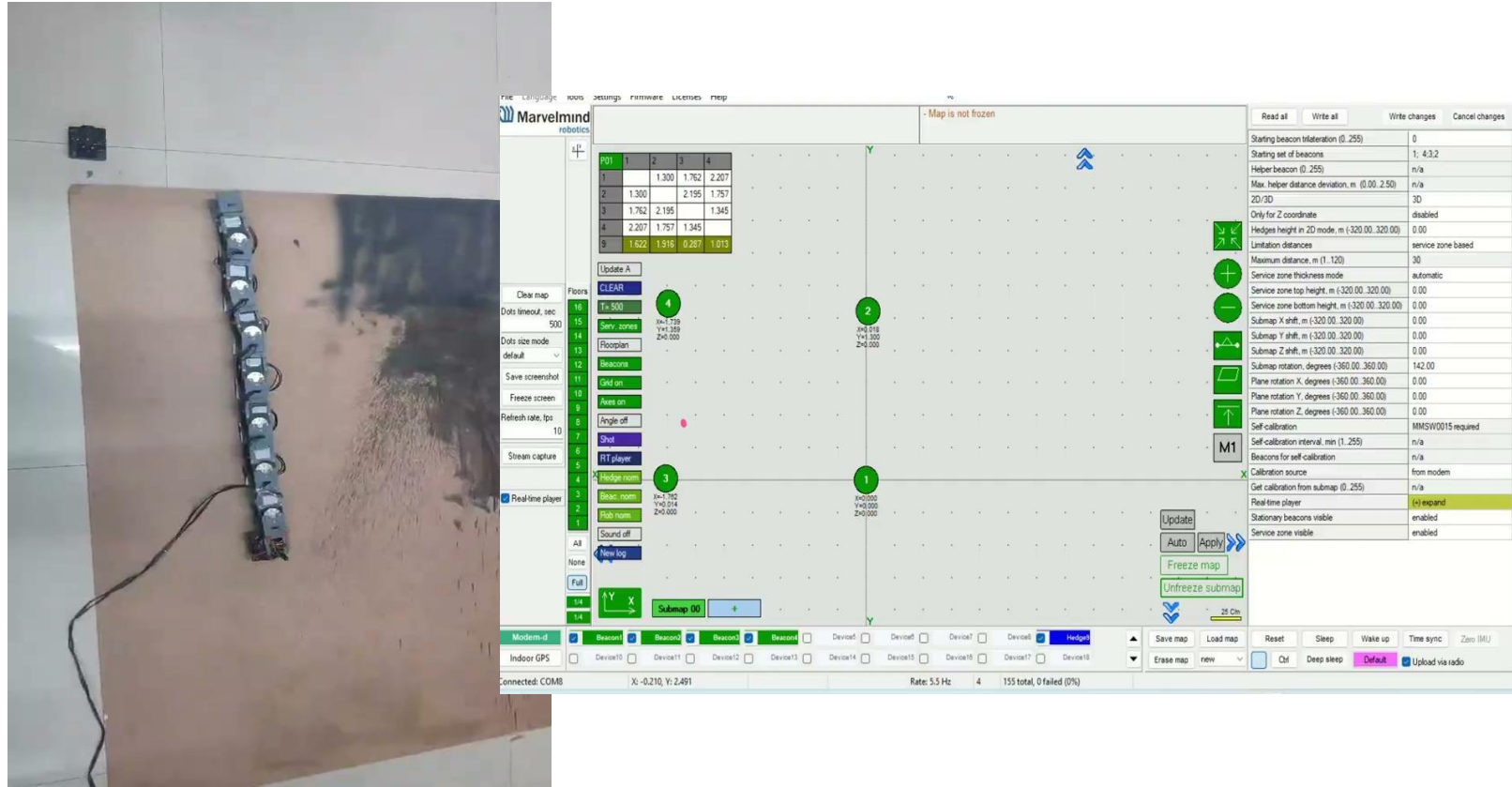
The governing equation for the i^{th} joint

$$\phi_{\text{ref}}^i = \begin{cases} \alpha_i \sin(\omega t + i\delta), & \text{if } i \text{ is even} \\ \alpha_i \sin(\omega t + 2i\delta), & \text{if } i \text{ is odd} \end{cases}, \quad i = 1, \dots, N - 1.$$

where N is the number of links, ϕ_{ref}^i is the reference joint-angle for the i^{th} motor, α_i the amplitude for i^{th} motor and ω is the temporal frequency. The motors for odd i have the axis parallel to the plane of the motion and the motors for even i have the axis perpendicular to the plane of the motion.

ROS simulation for Sidewinding.

Hardware implementation of lateral undulation



Lateral undulation motion and head movement observed using marvel mind sensors

Experimentation of lateral undulation gait on various surfaces



Artificial Grass



Door Mat



Mica sheet



Rough floor



Mat

Issues with lateral undulation

- Experimental motion is significantly slower than model predictions
- Reason: The viscous friction model used in the simulations can be given as

$$f_R = \begin{bmatrix} f_{R,x} \\ f_{R,y} \end{bmatrix} = - \begin{bmatrix} c_t (C_\theta)^2 + c_n (S_\theta)^2 & (c_t - c_n) S_\theta C_\theta \\ (c_t - c_n) S_\theta C_\theta & c_t (S_\theta)^2 + c_n (C_\theta)^2 \end{bmatrix} \begin{bmatrix} \dot{X} \\ \dot{Y} \end{bmatrix}$$

The propulsive force on a single link along x- direction can be given as

$$F_{\text{prop},i} = -F_x(\theta_i)\dot{x}_i - F_y(\theta_i)\dot{y}_i$$

where,

$$F_x(\theta_i) = c_t \cos^2(\theta_i) + c_n \sin^2(\theta_i)$$

$$F_y(\theta_i) = (c_t - c_n) \sin\theta_i \cos\theta_i$$

- c_t and c_n is the coefficient of friction in tangential and normal direction respectively.
- We had assumed c_t and c_n 1 and 10 respectively (for simulation purpose).
- However when we tested for our experimental environment c_t and c_n was found as-
 - a) Link without rubber grooves $c_t=0.291$ and $c_n=0.326$
 - b) Link with rubber grooves $c_t=0.821$ and $c_n=0.958$



Mat surface
texture

Rubber grip
below link

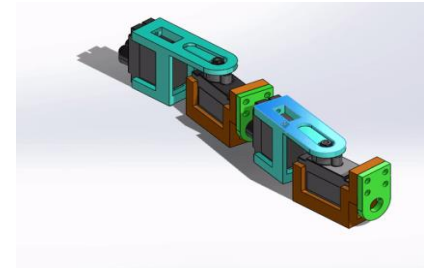
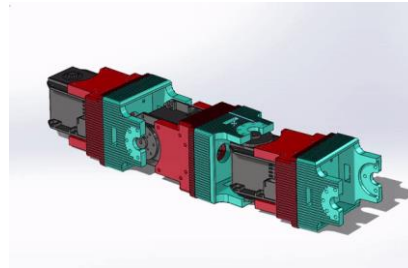
Issues with lateral undulation



- Biological snakes lift parts of their body while executing the lateral undulation mode of locomotion. Lifting helps in weight redistribution and hence enhance the net friction effectively to maneuver.

Design 2 and 3 will resolve this issue:

- Design snake model with an additional degree of freedom in the direction vertical to the plane on which it is moving.



Marvelmind super-MP-3D



Whole set up



Beacon



Modem

- Super -MP-3D sensor is used for precise ($\pm 2\text{cm}$) indoor positioning and navigation
- It can be used for autonomous indoor navigation of mobile robots, vehicles, and drones.
- It consists of 5 beacons and one modem.
- Beacons have **6D IMU** (3D gyro + 3D accelerometer) onboard.
- Beacon can work both **as stationary and as mobile**
- Modem is a central controller of the system and must be always active, when the system is running.
- Modem is used to set up the system, monitor it, and interact with the Dashboard (computer).
- Modem can be placed with all beacons in the radius of up to 100 meters.

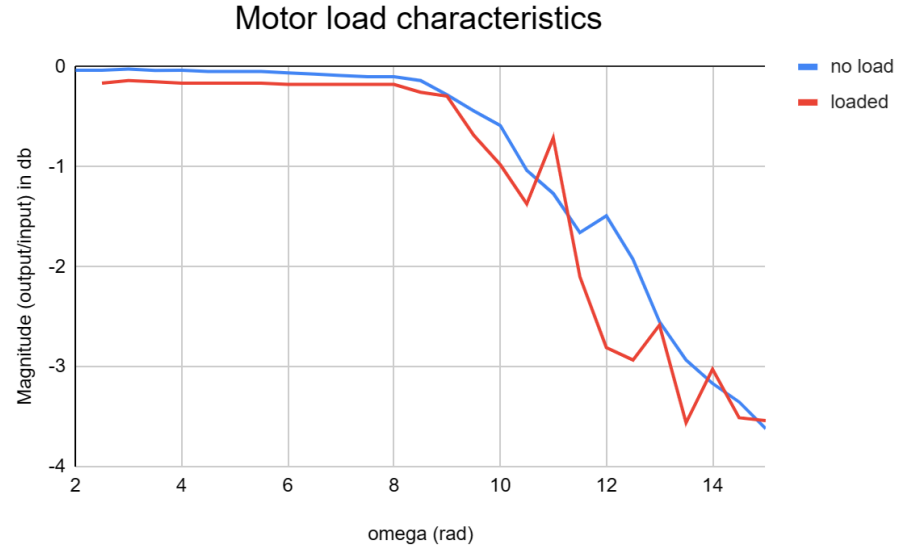
Load test of the motor

Weight suspended $m = 4.5 \text{ kg}$

Acceleration due to gravity $= 9.8 \text{ m/s}^2$

Radius of the horn, $r = 0.012 \text{ (m)}$, Torque ($\mathcal{T}$) $= rmg = 0.529 \text{ Nm}$

(approximately equal to lifting 3 links of the robot)



Band width of the motor is $14 \text{ rad/sec} = 2.38 \text{ Hz}$

Serpenoid robots



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- Intern: Devashish Sharma